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REVIEW ARTICLE

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# Acupuncture plus moxibustion for herpes zoster: A systematic review and meta-analysis of randomized controlled trials

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## Abstract

Herpes zoster is an acute inflammatory condition which can have a significant impact on quality of life. Antiviral therapies are effective, but do not meet patients' expectations of symptomatic relief. Acupuncture and moxibustion have been used for herpes zoster; this systematic review evaluated their efficacy and safety. Nine English and Chinese databases were searched from their inceptions to March 2016. Randomized controlled trials evaluating the combination of acupuncture plus moxibustion in adult herpes zoster were included. Outcomes included pain intensity and duration, quality of life and adverse events. Meta-analysis was performed using RevMan software (version 5.3). Nine studies (945 participants) were included. Studies were of low to moderate methodological quality based on risk of bias assessment. Pain intensity (visual analogue scale) was lower among those who received acupuncture plus moxibustion compared with pharmacotherapy (one study; MD −8.25 mm, 95% CI −12.36 to −4.14). The clinical significance of this result is yet to be established. Some benefits were seen for other pain and cutaneous outcomes, and global improvement in symptoms. Mild adverse events were reported in the intervention groups. Acupuncture plus moxibustion may improve pain and cutaneous outcomes, although current evidence is limited by the number of studies and methodological shortcomings.

KEYWORDS

acupuncture, herpes zoster, moxibustion, review, meta-analysis, systematic, varicella

## 1 | INTRODUCTION

Herpes zoster is the result of reactivation of the varicella zoster virus, usually acquired during childhood (Dworkin et al., 2007). The acute stage of herpes zoster is characterized by severe pain and unilateral lesions, with significant impact on the patient's quality of life (Gater et al., 2014; Johnson et al., 2010). Three clinical phases are seen: prodromal phase (the occurrence of pain before lesions erupt); acute phase (vesicles develop, postulate, and crust); and the healing phase (disappearance of lesions crusts) (Volpi, Gross, Hercogova, & Johnson, 2005). Lesions are typically limited to one dermatome, appear in clusters, and are accompanied by erythema (Dworkin et al., 2007; Gross et al., 2003).

As many as 90% of adults are at risk of developing herpes zoster (Ozaki, Miwata, Matsui, Kido, & Yamanishi, 1990). The global median

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incidence of herpes zoster was estimated at 4–4.5 per 1,000 person years (Yawn & Gilden, 2013). Age is a key risk factor for developing herpes zoster (Volpi et al., 2005), due to decreasing varicella zoster virus specific cell-mediated immunity with advancing age (Gershon et al., 2010). Other risk factors include Caucasian ethnicity (Gershon et al., 2010), female gender (Gershon et al., 2010), family history (Ansar, Farshchian, Ghasemzadeh, & Sobhan, 2014; Hicks et al., 2008), diabetes (Heymann et al., 2008), and mechanical trauma (Gershon et al., 2010).

Herpes zoster is usually self-limiting in immunocompetent patients (Volpi et al., 2005), with the rash resolving within 3–4 weeks (Carrasco, Straten, & Tyring, 2000). Management with antiviral therapies is recommended for patients over 50 years, immunocompromised patients, and patients with zoster involving more than one dermatome (Dworkin et al., 2007; Gross et al., 2003). Clinical practice guidelines recommend antiviral therapies to hasten healing time (acyclovir, valacyclovir, famciclovir, and brivudin) (Dworkin et al., 2007; Gross et al., 2003), and pain management treatments such as paracetamol/acetaminophen, opioid analgesics, and tricyclic antidepressants and anticonvulsants for significant pain (Dworkin et al., 2007; Volpi et al., 2005).

Antiviral therapies can reduce healing time, however a recent Cochrane review (Chen et al., 2014), found they were not effective in reducing the incidence of post-herpetic neuralgia (PHN), a common sequelae of herpes zoster. Gater et al. (2014) evaluated patient satisfaction with treatment for herpes zoster using the Treatment Satisfaction Questionnaire for Medication (TSQM, version II). Patient satisfaction was highest on the "side effects" domain and lowest on the "effectiveness" domain of the TSQM. The authors highlighted this as an area of unmet clinical need.

Acupuncture has been shown to be an effective treatment for some types of pain (Lee & Ernst, 2011), and various skin conditions (Chen & Yu, 2003), and may provide benefit during the acute stage of herpes zoster (Ursini et al., 2011). Moxibustion involves the burning of the herb *Artemesia vulgaris* L. close to the skin, usually near acupuncture points. A systematic review of moxibustion for pain found moxibustion reduced pain associated with osteoarthritis and may reduce pain in scleroma and herpes zoster (Lee, Choi, Kang, Lee, & Ernst, 2010). Acupuncture and moxibustion are recommended in Chinese medicine clinical practice guidelines for treating the acute symptoms of herpes zoster (Liu et al., 2013), and are frequently used in combination in clinical practice. This review examined the efficacy and safety of the combination of acupuncture with moxibustion in the acute stage of herpes zoster.

## 2 | METHODS

Five English (PubMed, Embase, CINAHL, CENTRAL, and AMED) and four Chinese (CBM, CNKI, CQVIP, Wanfang Data) databases were searched from inceptions to February 2014, with an update search run in March 2016. Search terms were grouped according to condition (herpes zoster, zoster, varicella zoster virus, and variants), intervention (acupuncture, moxibustion, and variants), and study design (randomized, controlled, placebo, and variants).

Resulting citations were exported to reference management software for review. Titles and abstracts were screened to identify relevant citations, and full text retrieved where further review was required. Studies were eligible if they: (a) were randomized controlled trials, (b) included adults (≥18 years) with acute stage herpes zoster, (c) evaluated the combination of acupuncture plus moxibustion, and (d) used guideline recommended treatments (antiviral therapy and/or analgesics) (Dworkin et al., 2007; Gross et al., 2003; Volpi et al., 2005), no treatment or sham/placebo as the comparator. As diagnosis is usually clinical, no additional requirements for diagnosis were applied. Studies which evaluated the combination of acupuncture plus moxibustion alone, or as adjunct to guideline recommended treatments were eligible.

Studies were excluded if the condition being treated was PHN, complications of herpes zoster (Ramsay Hunt syndrome, zoster opthalmicus, zoster sine herpete, visceral or disseminated zoster, bacterial infections), or included immunocompromised patients (eg HIV, cancer, diabetes, pregnant, or breastfeeding). Studies where the comparator was herpes zoster vaccine or a Chinese medicine therapy were also excluded.

The primary outcomes were pain intensity, measured on visual analogue scale (VAS), McGill pain score, or other rating scales (e. g., verbal rating scale), and time to resolution of pain. Secondary outcomes included measures of lesion healing (time to resolution of rash, time to cessation of new lesion formation, time to crust formation); health related quality of life (HRQoL), measured on the Zoster Brief Pain Inventory, Initial Zoster Impact Questionnaire, or other quality of life measures; incidence of PHN; therapeutic effective rate (TER), defined as a lesion improvement by 30% or more, and significant reduction in pain, according to a Chinese guideline (State Administration of Traditional Chinese Medicine, 1994).

Characteristics of studies, details of interventions and comparators, outcomes and results were extracted into a pre-defined spreadsheet. Quality assessment was made using the Cochrane Collaboration's risk of bias tool independently by two researchers (KW, WZ). If disagreement was seen in assessments, this was resolved through discussion with a third researcher (MC).

Data were analysed using RevMan software (The Nordic Cochrane Centre) Continuous data were presented as mean difference (MD) and dichotomous data were presented as risk ratio (RR), with 95% confidence intervals (CI). A random effects model was used for all analyses. Statistical heterogeneity was considered present if $p < .10$ in the Chi square test for heterogeneity, and the $I^2$ statistic was greater than 50%. Statistical heterogeneity was explored through sensitivity analysis for studies assessed as low risk of bias for sequence generation, and further sub-group analysis was planned according to comparator type and use of local or distal acupuncture points. If 10 or more studies were included in a meta-analysis, we planned to explore publication bias through visual inspection of funnel plots. This was not possible, as none of the meta-analyses included 10 studies. The protocol for this review was published in PROSPERO (Registration no. CRD42015029303).

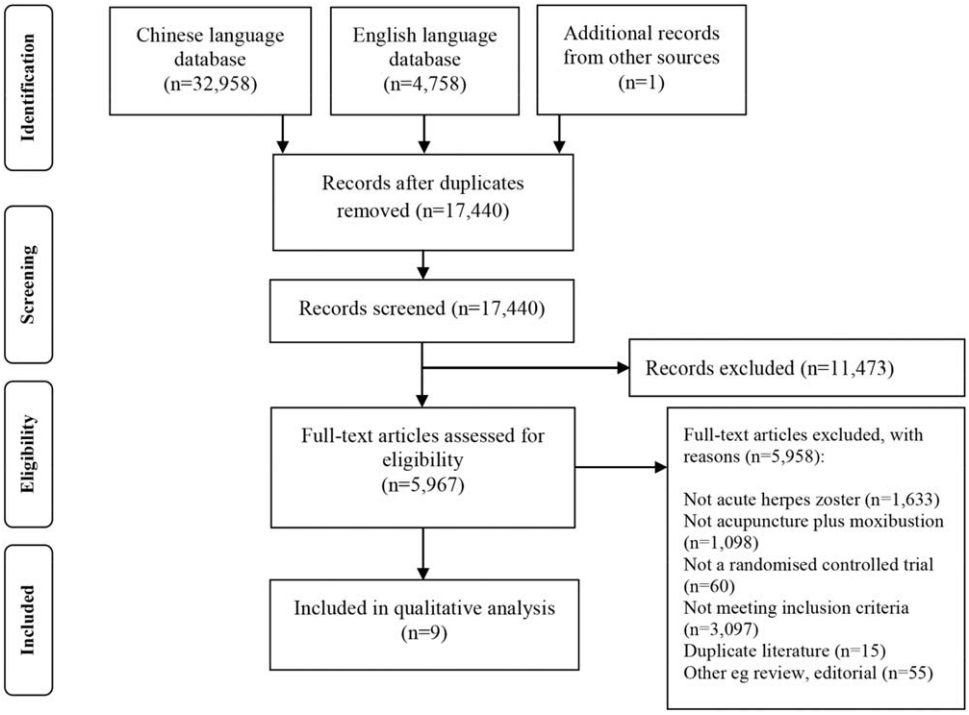


**FIGURE 1** Study selection flow chart

## 3 | RESULTS

Database searches identified more than 37,000 citations (Figure 1). More than 11,000 were excluded based on screening of titles and abstracts, and the full text of almost 6,000 studies were reviewed. Nine randomized controlled trials (945 participants) met the inclusion criteria (Table 1) (Bao, 2011; Chen, 2012; Li, 2011; Li, Gou, Zhang, & Wang, 2014; Lin & Zhao, 2012; Liu, 2010; Lu, 2004; Wang, 2011; Yang, 2013). All studies were conducted in China, and compared the combination of acupuncture plus moxibustion with pharmacotherapy. Two studies included more than one treatment arm, and those additional treatment arms were excluded from this review (Li, 2011; Liu, 2010).

The sample size of included studies ranged from 56 (Lu, 2004) to 400 participants (Li, 2011), and the median sample size was 62. Participants were recruited from hospital outpatient departments (Li, 2014; Lin & Zhao, 2012; Liu, 2010; Wang, 2011; Yang, 2014), inpatient departments (Chen, 2012), or both inpatient and outpatient departments (Li, 2011; Liu, 2010; Yang, 2014). The duration of herpes zoster ranged from one (Li, 2011; Li et al., 2014; Liu, 2010; Yang, 2014) to 10 days (Bao, 2011; Lu, 2004). Treatment was provided for two weeks in one study (Yang, 2014), and for 10 days in remaining studies. Four studies reported follow-up assessment, which was made at 30 days (Lin & Zhao, 2012), 60 days (Li, 2011), or 90 days/three months (Liu, 2010; Lu, 2004). Two studies lost participants to follow up ($n = 32$) (Li, 2011; Wang, 2011).

### 3.1 | Intervention and comparator

All studies evaluated the combination of acupuncture plus moxibustion against pharmacotherapy. Studies used various acupuncture approaches, including manual acupuncture (Bao, 2011; Chen, 2012; Li et al., 2014; Lin & Zhao, 2012; Yang, 2014), inserting needles surrounding the lesion sites ("surrounding acupuncture", SA) (Bao, 2011; Li, 2011; Lin & Zhao, 2012; Lu, 2004), and electroacupuncture (Bao, 2011; Li, 2011; Lin & Zhao, 2012; Lu, 2004). As all approaches are typical of acupuncture clinical practice, it was considered suitable to pool results for analysis. The most commonly used acupuncture approach was to treat the rash area (five studies) (Chen, 2012; Lin & Zhao, 2012; Lu, 2004; Wang, 2011; Yang, 2014) or to use *ashi* points (local tender points, or points surrounding the rash area; four studies) (Li, 2011; Li et al., 2014; Liu, 2010; Wang, 2011) or *Huatuojiaji* points (points parallel to the spine; four studies) (Li, 2011; Li et al., 2014; Liu, 2010; Wang, 2011). Other commonly reported acupuncture points were GB34 *Yanglingquan* (Bao, 2011; Lin & Zhao, 2012) and LR3 *Taichong* (two studies each) (Bao, 2011; Lin & Zhao, 2012).

Treatments used in the comparator groups included guideline recommended treatments, alone or in combination with other treatments. Two studies compared acupuncture plus moxibustion to antiviral therapy alone (acyclovir) (Chen, 2012; Lin & Zhao, 2012), three studies used antiviral therapy such as acyclovir (Bao, 2011) and valacylovir (Li, 2011; Li et al., 2014) in combination with vitamins (B1, B12, or derivatives) (Bao, 2011; Li et al., 2014) or vitamins and saline compresses (Li, 2011). One study used the antiviral drug moroxydine in combination with the anticonvulsant carbamazepine (Lu, 2004), and three studies used antiviral drugs acyclovir (Yang, 2014) or valacyclovir (Liu, 2010; Wang, 2011) with pain medications gabapentin (Yang, 2014), prednisone (Wang, 2011), or carbamazepine (Liu, 2010) and other treatments (Liu, 2010; Wang, 2011; Yang, 2014). Vitamins B1 and B12 are commonly used in China to aid nerve repair. Of the nine studies, only one (Lin & Zhao, 2012) used antiviral therapy at doses recommended in clinical practice guidelines (Dworkin et al., 2007).

**TABLE 1** Characteristics and intervention details of the included trials

| First author, publication year | Treatment duration, follow-up duration | Duration of condition (mean or range) (days) | No. of participants randomised/ assessed, dropouts | Age (mean ±SD) or range) | Acupuncture points | Comparator |
|---|---|---|---|---|---|---|
| Bao (2011) | 10 days, NS | I: 2∼10<br>C: 2∼8 | I: 32/32, 0<br>C: 32/32, 0 | I: 51<br>C: 53 | MA: LI11 Quchi, TE5 Waiguan, SP6 Sanyinjiao, SP10 Xuehai, ST36 Zusanli, LR14 Qimen, LR3 Taichong, GB34 Yanglingquan<br>SA: rash area<br>Moxa: rash area | Acyclovir 0.8g, tid, po; mecobalamin, im, qod |
| Chen (2012) | 10 days, NS | I: 3∼6<br>C: 2∼5 | I: 31/31, 0<br>C: 31/31, 0 | I: 38∼72<br>NS | Rash area (no further details provided) | Acyclovir 0.25g, bid, iv; acyclovir ointment, ext. |
| Li et al. (2014) | 10 days, NS | Total: 1-7 | I: 32/32, 0<br>C: 30/30, 0 | I: 34.1<br>C: 35.2 | MA: Huatuojiaji, ashi points<br>EA: Huatuojiaji<br>Moxa: ashi points | Valacyclovir 0.3g, bid, po; vitamin B1 po tid; vitamin B12 po tid |
| Li (2011) | 10 days, 60 days | I: 5.56<br>C: 5.26 | I: 100/94, 6<br>C: 100/95, 5 | I: 47.54<br>C: 46.42 | SA: ashi points<br>EA: Huatuojiaji, TE6 Zhigou, SI3 Houxi<br>Moxa: ashi points | Valacyclovir 0.3g, bid, po; vitamin B1 po tid; saline compress |
| Lin and Zhao (2012) | 10 days, 30 days | I: 4.37<br>C: 4.54 | I: 32/32, 0<br>C: 30/30, 0 | I: 53.26<br>C: 52.42 | MA: LI4, LR3, Taiyang, TE6, GB34<br>SA: rash area<br>Moxa: rash area | Acyclovir 0.8g, 5 times per day po; acyclovir cream, 5 times per day, ext. |
| Liu (2010) | 10 days, 90 days | I: 4.96<br>C: 5.03 | I: 30/30, 0<br>C: 30/30, 0 | I: 47.91<br>C: 47.79 | EA: ashi points, Huatuojiaji<br>Moxa: ashi points | Valacyclovir 0.3g, bid, po; carbamazepine 0.1g, tid, po; vitamin B1 po tid |
| Lu (2004) | 10 days, 3 months | I: 3-10<br>C: 4-9 | I: 28/28, 0<br>C: 28/28, 0 | I: 55∼73<br>C: 56∼68 | SA: rash area<br>Moxa: rash area | Carbamazepine 0.1g, qd, po; Moroxydine 0.6g, qd, po |
| Wang (2011) | 10 days, NS | Total: <7 | I: 64/58, 6<br>C: 56/56, 0 | I: 63.5<br>C: 62.3 | EA: Huatuojiaji, ashi points<br>Moxa: rash area | Valacyclovir 0.3g, bid, po; prednisone 10mg, tid, po; acyclovir cream, 6 times per day, ext. |
| Yang (2014) | 14 days, NS | I: 5.13<br>C: 4.77 | I: 29/292, 0<br>C: 30/30, 0 | I: 38.79<br>C: 41.97 | Rash area (no further details provided) | Acyclovir 0.4g, tid, po; gabapentin 0.3g, tid, po; acyclovir ointment, ext.; mecobalamin po tid |

bid, twice a day; C, control group; EA, electroacupuncture; ext., external use; I, intervention group; im, intramuscular; m, mean; MA, manual acupuncture; Moxa, moxibustion; NS, not specified; po, oral use; qd, once a day; qod, every other day; SA, surrounding acupuncture; SD, standard deviation; tid, three times a day.

### 3.2 | Risk of bias assessment

Three studies used a random number generator (Lin & Zhao, 2012; Liu, 2010; Wang, 2011) and one used computer based central randomization (Li, 2011); all were assessed as low risk of bias for sequence generation. The remaining studies did not provide sufficient information about randomization processes, and were judged to be at unclear risk of bias. One study which used central allocation was judged as low risk of bias for allocation concealment (Li, 2011), while all others were judged as unclear risk due to insufficient detail. Due to the nature of the intervention and comparator, it was not possible to blind participants and personnel to group allocation, and all studies were judged to be at high risk of bias. One study reported blinding outcome assessors to group allocation (Li, 2011), and was considered to be at low risk of bias for this domain. One study reported participant loss but did not describe the reasons (Li, 2011). As the number of dropouts was similar in both groups, this study was judged as low risk of bias. Another study reported loss to follow up (Wang, 2011) ($n = 6$), but the number in each group was not specified. This study was judged as unclear risk of bias. Other studies reported data for all participants randomized and were judged as low risk. One trial was registered (Li, 2011), but the details provided were insufficient, and the study was considered as unclear risk of bias for selective outcome reporting. No trial protocol or registration was available for all other studies, which were at unclear risk of bias.

## 4 | PAIN OUTCOMES

### 4.1 | Pain intensity

One study reported on VAS pain scores at the end of treatment and at 60 days (Li, 2011). Pain score at end of treatment was 8.25 mm lower in those who received acupuncture plus moxibustion compared to

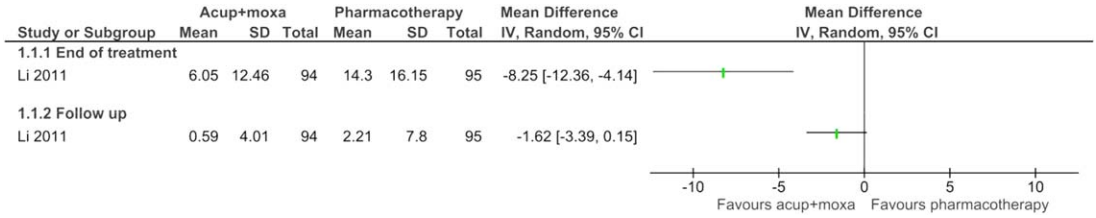


**FIGURE 2** Pain score. Notes: Acup: acupuncture; CI: confidence interval; moxa: moxibustion; SD: standard deviation

valacyclovir plus vitamin B1 and saline compress, with significant statistical difference shown (MD −8.25 mm; 95% CI −12.36 to −4.14) (Figure 2). Whilst statistically significant, this result is less than a 1cm difference between groups. This is unlikely to be clinically meaningful, particularly when considering the wide variation in results and the trial being unblinded. This effect was not sustained at follow up (MD −1.62 mm; 95% CI −3.39 to 0.15).

### 4.2 | Time to resolution of pain

Two studies reported time to resolution of pain (Lin & Zhao, 2012; Wang, 2011). One study did not specify the time from which measurement was made (Wang, 2011), and data were excluded from this outcome. The other study reported the time from start of treatment to resolution of pain (Lin & Zhao, 2012). Pain resolution was achieved 6.59 days earlier in the participants who received acupuncture plus moxibustion compared to oral and topical acyclovir (MD −6.59 days; 95% CI −8.07 to −5.11). It should be noted that topical acyclovir is not effective for herpes zoster and is not recommended in clinical practice guidelines (Dworkin et al., 2007).

## 5 | CUTANEOUS OUTCOMES

### 5.1 | Time to resolution of rash

Three studies reported on time to resolution of rash (Li, 2011; Li et al., 2014; Wang, 2011). Two studies did not specify the time point from which measurement was made, and were not analysed (Li et al., 2014; Wang, 2011). One study reported the time from rash onset to resolution (Li, 2011). Resolution of the rash occurred 3.4 days earlier in the acupuncture plus moxibustion group than in the comparator group (valacyclovir plus vitamin B1 and saline compress; MD −3.4 days; 95% CI −3.71 to −3.09).

### 5.2 | Time to crust formation

Five studies reported on time to crust formation (Li, 2011; Li et al., 2014; Lin & Zhao, 2012; Liu, 2010; Wang, 2011). Three studies did not specify the time point from which measurements were made, and were excluded from the analysis (Li et al., 2014; Liu, 2010; Wang, 2011). Crust formation measured from rash onset occurred 1.42 days earlier in those who received acupuncture plus moxibustion compared with those who received valacyclovir plus vitamin B1 and saline compress (MD −1.42 days; 95% CI −1.52 to −1.32) (Li, 2011). When measured from the start of treatment, benefit was also seen with acupuncture plus moxibustion compared with oral and topical acyclovir (Lin & Zhao, 2010), with crust formation occurring 1.64 days earlier (MD −1.64 days; 95% CI −2.87 to −0.41).

### 5.3 | Time to cessation of new lesion formation

Four studies reported the time to cessation of new lesion formation (Li, 2011; Li et al., 2014; Lin & Zhao, 2012; Liu, 2010), with two excluded from analysis due to lack of detail about the time from which measurements were made (Li et al., 2014; Liu, 2010). Acupuncture plus moxibustion resulted in a lesser time to cessation of new lesion formation compared with valacyclovir, vitamin B1 and saline compress when measured from rash onset (MD −0.29 days; 95% CI −0.35 to −0.23) (Li, 2011). When measured from start of treatment, acupuncture plus moxibustion reduced the time to cessation of new lesions compared with oral and topical acyclovir (MD −1.26 days; 95% CI −2.16 to −0.36) (Lin & Zhao, 2012).

#### 5.3.1 | Incidence of PHN

Four studies reported on the incidence of PHN (Li, 2011; Lin & Zhao, 2012; Liu, 2010; Lu, 2004). Sub-group analysis was performed according to PHN definition (one/three months after resolution of the rash). Meta-analysis showed a significant reduction in the incidence of PHN (one month after rash resolution) in those who received acupuncture plus moxibustion compared to pharmacotherapy (two studies; RR 0.29; 95% CI 0.16 to 0.53; $I^2 = 0\%$) (Li, 2011; Lin & Zhao, 2012). In studies which measured PHN three months after rash resolution, no statistical difference was seen (two studies; RR 0.16; 95% CI 0.02 to 1.35; $I^2 = 0\%$) (Liu, 2010; Lu, 2004).

#### 5.3.2 | Therapeutic effective rate

Of the six studies which reported therapeutic effect rate (Bao, 2011; Chen, 2012; Lin & Zhao, 2012; Liu, 2010; Wang, 2011; Yang, 2014), one study reported data in a different way to that described in the guideline (Yang, 2014). This study was excluded from analysis. The chance of achieving a 30% or greater improvement in lesions and significant pain reduction with acupuncture plus moxibustion was 2.67 that of pharmacotherapy (RR 2.67; 95% CI 2.03 to 3.52; $I^2 = 43\%$).

#### 5.3.3 | Adverse events

Two studies reported on adverse events (Li, 2011; Liu, 2010). One of them reported no adverse events occurred (Liu, 2010). Two cases of haematoma and five cases of bleeding were reported in the

intervention group of one study (Li, 2011). The adverse events were mild and the symptoms were relieved by local pressing or massage. No adverse events were reported in any of the control groups. Seven studies did not report any information on adverse events.

## 6 | DISCUSSION

The findings from this systematic review suggest potential benefit from acupuncture and moxibustion in reducing pain intensity, improving rash healing, and reducing the incidence of PHN, although few studies reported these outcomes. Meta-analysis of five studies showed improvement in lesions and significant reduction in pain based on the therapeutic effective rate. Therapeutic effective rate has not been validated, and should be interpreted with caution.

Acupuncture and moxibustion were well tolerated by patients, with few mild side effects reported. Bleeding and hematoma are the most common adverse events with acupuncture (Park, Lee, Choi, Kim, & Choi, 2010; White, Hayhoe, Hart, & Ernst, 2001; Witt et al., 2011), resulting from needles puncturing the capillary vessels (Liang, 2005). Due care minimizes the risk of this occurring. Although no adverse events were reported related to moxibustion in the included studies, appropriate distance between burning moxibustion cigar and acupuncture points should be maintained to avoid empyrosis (Liang, 2005).

The findings in relation to incidence of PHN require further investigation. There is no consensus on the definition of PHN, but pain persisting three months after the resolution of the rash is generally accepted as the clinical definition and is used for research purposes (Dubinsky, Kabbani, El-Chami, Boutwell, & Ali, 2004). The differing findings seen when PHN was assessed at one month and three months may relate to the definition used. Pain from acute stage herpes zoster has been reported to take up to 61 days to resolve (Ahn et al., 2001). In this review, fifty cases of pain were reported one month after rash resolution, while six cases (all in the comparator group) were reported three months after resolution of the rash. This follows the trend for zoster related pain to resolve over time, and is not likely to be an indicator of the efficacy of acupuncture plus moxibustion.

The pathogenesis of acute pain in herpes zoster is the inflammatory response and necrosis of the varicella zoster virus infected sensory nerve (Dworkin et al., 2007). Acupuncture and moxibustion are both important analgesic therapies in the acute stage of herpes zoster (Liu et al., 2013). The analgesic effect of acupuncture has been well-researched. Acupuncture promoted cortical endocrine involvement through evoking endogenous opiates, such as β-endorphin, encephalin, dynorphin, and endomorphin (Lin & Chen, 2008). Other research suggests acupuncture could produce anti-inflammatory effects via the autonomic nerve system (ANS) in animal models (Sekido, Ishimaru, & Sakita, 2003). The mechanism of the anti-nociception effect of electro-acupuncture is still unclear. Recent animal studies showed that electro-acupuncture may alleviate inflammation by suppressing certain pain-related molecular, protein and mRNA pathways (Du, Fang, Liang, & Fang, 2014; Fang, Du, Liang, & Fang, 2013; Liang, Fang, Du, & Fang, 2012).

Antiviral therapy (with or without analgesic therapy) is recommended as first line therapy for immunocompetent patients over 50 years of age (Dworkin et al., 2007; Volpi et al., 2005; Gross et al., 2003). Recently released guidelines suggest antiviral therapy may be initiated in patients under 50 years (Werner et al., 2017). The recommended dose of antiviral drug valacyclovir is one gram three times daily for seven days, and for oral acyclovir is 0.8 g five times a day for seven days (Gross et al., 2003). Only one study included in this review used the guideline recommended dosage (Lin & Zhao, 2012), while the dosage of the other studies included in this review was considerably less than those recommended in guidelines. When acupuncture plus moxibustion was compared with guideline recommended dose of antiviral therapy, benefits were seen in the time to resolution of pain, time to crust formation, and time to cessation of new lesions. The study also used a random number for group allocation, reducing bias from sequence generation. In light of this, this study may provide the best evidence for treatment effects of acupuncture plus moxibustion.

Achieving adequate analgesia is a key aim of pharmacological management (Dworkin et al., 2007). Recent guidelines recommend analgesic treatment follow the WHO pain ladder (Werner et al., 2017). Titration can be considered if analgesia is not achieved (Dworkin et al., 2007), or basic analgesia can be supplemented with tricyclic antidepressants or anticonvulsants (Werner et al., 2017). Hence, clinical practice uses an individualized approach to pain management. As standard doses of pain management medications were used in included studies, it is possible that the findings are not directly translatable to clinical practice. Two studies used the anticonvulsant carbamazepine as analgesic therapy for managing acute zoster pain (0.1 g per day) (Liu, 2010; Lu, 2004), which was also lower than the guideline recommended dosage (0.4–1.2 g per day) (Gross et al., 2003). None of the included studies described referring to clinical practice guidelines, and no Chinese guidelines for conventional medical management of herpes zoster could be identified to verify the dosage of antiviral and anticonvulsant therapies. Finally, several studies (Chen, 2012; Lin & Zhao, 2012; Wang, 2011; Yang, 2014) used topical application of antiviral therapy, which is contrary to clinical practice guidelines. Therefore, the clinical relevance of the findings of this review are uncertain.

Several studies included in this review used vitamin B medications or derivatives, which are not recommended in clinical practice guidelines. In China, the prescribing practice for acute stage of herpes zoster often combines vitamin B1 and B12 for repairing infected nerves and regulating the function of synaptic transmission (Chen, Wu, & Jiang, 2006; Zhang & Huang, 2007). It is unclear whether the addition of vitamin B1 or B12 alters the efficacy of valacyclovir, and results should be interpreted in light of this.

Many of the studies included in this review had methodological flaws which limits interpretation of the results. There was a lack of detail about the randomization process, blinding and outcome assessment. These factors are likely to influence the reliability of the results. With the exception of the VAS, few outcomes in herpes zoster research have been standardised or validated. With a complex intervention like acupuncture, the selection of objective outcomes becomes even more important to minimise bias in the results.

## 7 | CONCLUSION

Several studies have evaluated the combination of acupuncture plus moxibustion in the acute stage of herpes zoster. While single studies have produced some positive results, overall there is insufficient evidence from high quality studies reporting on objective outcomes. The lack of detail relating to some outcomes such as time to resolution of pain or rash limited evaluation of the evidence. Further research comparing acupuncture plus moxibustion to guideline recommended doses of pharmacotherapy, which report definitions used for outcomes and which use the accepted definition for PHN are needed.

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### CONFLICT OF INTEREST

No conflict of interest.

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